

Index

Note: Page numbers followed by *f* indicate figures, *t* indicate tables, *b* indicate boxes and *np* indicate footnotes.

A

- A330 airbus model, 606–607
- ABC News Nightline, 1999, 299–300, 325, 427–428
- Abernethy, C. N., 832
- Abstraction, 39–40, 57*b*, 285*b*, 289*b*, 597*b*
 - in ideation, 310
- Abstract work activity, 284*b*, 285, 299*b*
 - top-down design approach, 285–286
 - in architecture, 291*b*
 - characteristics, 289–290
 - disruptive design, 291
 - goals, 286–289
 - hedging against risks, 291
 - transition for, 290–291
 - as work kernel, 286*f*
- Acceptance test, 634–635
- Acohido, B., 738
- Action-object approach, 195
- Active entities, social model, 216–217
- Active learning, 39
- Activity-based interaction, 18–19, 19*b*, 21, 62*b*, 139, 306, 306*b*, 352*b*
- Activity theory, 252–253
 - prototyping, 407*b*
- Adams, D., 823–824
- Adler, C., 59
- Affinity diagram, 40, 154*b*, 155–156, 598–599.
 - See also* Work activity affinity diagram (WAAD)
- Affordances
 - cognitive (*see* Cognitive affordances)
 - design checklist, 689
 - emotional (*see* Emotional affordances)
 - functional (*see* Functional affordances)
 - Gibson's ecological view, 655, 665*b*
 - physical (*see* Physical affordances)
 - roles, 686–687
 - sensory (*see* Sensory affordances)
 - for sort capability, 687–689*b*
 - user-created affordances, 689–693
- UX design, 825–827
 - checklist, 687–689, 687*f*
 - cognitive, 828–830
 - concepts, 651
 - confusion over HCI/UX, 827–828
 - demystifying, 652
 - functional, 830–831
 - Gibson's ecological view, 830–831
 - issues, 652
 - types, 652–654, 653*f*
- After Scenario Questionnaire (ASQ), 519
- Agile lifecycle processes
 - chunking, 71–72, 71*f*
 - defined, 70
 - scope, 71–72, 71*f*
 - user-facing prototypes, 72–73
- Agile SE sprints, 76–77, 77*f*
- Agile software engineering (SE), 70*b*, 625*b*
 - abstraction, 631*f*
 - basics of, 627–630
 - big design upfront, avoidance, 630
 - challenges, 635–636
 - characteristics, 629–630
 - colocation, 636–637
 - customer stories, 632*f*
 - design and implementation, 637, 637*f*
 - early funnel, 636*b*, 640*b*

Agile software engineering (SE) (*Continued*)

- goals, 628
 - late funnel, 636*b*
 - lifecycle process approach
 - planning, 631–633
 - sprints, 634–635, 634*f*
 - operating in silos, 629
 - principles, 628
 - problems to anticipate, 637–638
 - scope *vs.* time, 631*f*
 - small upfront analysis and design, 640–641
 - synthesized approach, 638–647
 - UX process lifecycle, 636
 - integration of, 639–641
 - software side *vs.*, 638
 - waterfall lifecycle development methodology, 628–629
 - workflow synchronization, 643*f*
 - communication during, 646–647
 - continuous delivery, 646
 - dove-tailed work activities, 643–645
 - early delivery value, 645–646
 - planning across iterations, 646–647
 - regression testing, 646
- Agile UX funnel model, 123, 123*b*, 128*b*, 236*b*, 240*b*, 276*b*, 298*b*, 408–409*b*, 636*b*, 640*b*, 641, 644*t*
- large-scope approach, 485
 - small-scope approach, 486
- Agile UX method
- funnel model, 75*f*, 625, 625*b*
 - design creation sublifecycle, 277, 277*f*
 - early funnel activities, 78–80
 - late funnel activities, 76–77
 - need for, 74–75
 - scope in, 75
 - speed and rigor, 75–76
 - lifecycle process, 46*b*
 - mapping project parameters, 48*f*
 - synthesized approach, 638–647
- Alben, L., 831
- Albers, J., 821
- Albert, B., 522, 618
- Alpha testing, 449–450, 615*b*

- Alvarez, H., 818–819, 822
- Ambient intelligence, 104
- Ambient interaction, 102–104
- Analytic UX evaluation, 438*b*, 441–442, 444–445, 557*b*
- characteristics, 546–547
 - criticism, 547
 - design reviews, 547*b*, 548–549
 - design walk-throughs, 547
 - ecological-layer inspection, 559
 - emotional-layer inspection, 559–560
 - expert inspection
 - defined, 550
 - inspectors, 550–551
 - focus groups, 549
 - heuristic evaluation, 547*b*
 - interaction-layer inspection, 559
 - process speed, 545
 - pyramid of user needs, 559*b*
- Andersen, H., 254
- Andre, T. S., 441–442, 444
- Andres, C., 231
- Animated prototype, 429–430
- Ann, E., 12
- Anonymity, of participants in UX evaluation, 590–591
- Anthropomorphism, 815–818
- Antle, A. N., 105
- Architecture, 263
- Ardito, C., 12
- Arnold, A. G., 615
- Arnowitz, J., 74, 78
- Artifact model, 54*b*, 123*b*, 170, 179, 179*b*, 241*b*, 306–307, 307*b*. *See also* Interaction artifact model
- auto repair shop, 209–210*b*
 - construction, 208–210
 - defined, 207–208
 - information flow and, 136
 - restaurant flow model, 209*b*, 210*f*
 - of work practice, 135–136
- Assessment guidelines, UX design, 790*f*
- content and meaning of feedback, 795–804
 - existence of feedback, 790–792
 - information displays, 804–807

- presentation of feedback, 792–795
 - system feedback, 789–790
 - system response, 789
 - Assessment, stage of Interaction Cycle, 789–807
 - AttrakDiff questionnaires, 514, 524–526, 524–525*b*
 - Audiences report, 592–595
 - Auditory sensory design issues, 681–682
 - Automatic UX evaluation, 450–451
 - Automation, 786–789
 - Awkwardness, in physical interaction, 667, 779–780
- B**
- Back, M., 46
 - Baecker, R. M., 315
 - Bafna, P., 103–104, 313–314
 - Baker, L., 633, 635–636
 - Balli, U., 103–104, 313–314
 - Bangor, A., 519, 521, 563
 - Barnard, P., 98–99
 - Barton, S. B., 659–661
 - Basili, V., 555–556
 - Baskinger, M., 105, 312–313, 316
 - Bastien, J. M. C., 711
 - Baty, S., 295
 - Beale, R., 350–351
 - Beck, K., 71, 71*f*, 231, 627, 631*f*, 633
 - Becker, K., 737–738
 - Bekker, M. M., 603
 - Bell, T. E., 110–111
 - Bellotti, V., 254
 - Benchmark task, 423*b*, 448*b*, 488
 - empirical UX evaluation
 - crafting, 465–468
 - defined, 460
 - mechanics, 469–471
 - selection of, 460–465
 - error recovery, 463
 - rubrics, 468
 - task scripts, 469
 - usage-centered wording, 467
 - Benington, H. D., 111
 - Bennett, J. L., 254, 455–456, 606
 - Bera, P., 821–822
 - Berry, R. E., 831
 - Besselaar, P. V. d., 398
 - Beta testing, 449–450, 615*b*
 - Beu, A., 11, 109–110, 525
 - Bevan, N., 28, 604, 711
 - Beyer, H., 187, 199, 231–232, 262, 281–282, 635–636, 639
 - Bias, R. G., 24
 - Biases, 282–284
 - Biers, D., 512
 - Billingsley, P. A., 832
 - Binomial probability distribution, 609, 609*f*
 - Biometrics, 618
 - Bittner, K., 256
 - Bjerknes, G., 252, 407
 - Blackberry smartphone, 283
 - Black & Decker Snakelight, 18*f*
 - Bloomberg UX evaluation lab, 494–496*b*, 494–496*f*
 - Blythin, S., 254
 - BMW iDrive, 783
 - Bødker, S., 84, 107, 252, 398–399
 - Boehm, B. W., 271
 - Boff, K. R., 831
 - Bolchini, D., 429
 - Boling, E., 308
 - Borchers, J., 362, 382–383, 428
 - Borman, L., 96
 - Bottom-up UX design approach
 - biases and constraints, roles, 282–284
 - existing work practice, designing for, 279–284
 - human-centered/user-centered design, 281–282
 - innovative possibilities, 284
 - Bowden, R., 28
 - Bradley, M. M., 526
 - Brainstorming, 43, 296
 - Branscomb, L. M., 95
 - Breemen, A. v., 104
 - Brisben, A., 104
 - Brooke, J., 517
 - Brown, C. M., 308, 831
 - Brown, L., 832
 - Brown, T., 23, 35, 85

Buchenau, M., 6, 430, 532

Building systems

- changes during project
 - external changes, 64
 - project requirements and parameters, 63–64
 - responding to, 65
- closing the gap, 65
- communication problems, 66–67
- designer's understanding, 65
- gap between views, 65
- perceived requirements, 66
- reality, 64

Burmester, M., 11, 109–110, 524–525

Bush, R. M., 831

Business case, for UX, 23–25

Butler, S., 604

Buur, J., 84

Buxton, B., 12, 35–36, 43, 101, 271, 314–315, 325, 369–371

C

Campbell, R., 573

Capra, M. G., 568

Card, S. K., 99, 108

Card sorting, 37, 40–41, 155

Carey, L., 438

Car heater control, 760*b*, 760*f*

Carmel, E., 397–398

Carroll, J. M., 110, 253, 270, 330–331, 338, 438, 442, 615

Carter, P., 450

Casey, M. A., 126

Castillo, J. C., 450, 508–509, 615

Catalysts, ideation, 298

- eureka moment, 308–310

Chain of custody, 180

Cheok, A., 652

Chin, J. P., 515

Chrissis, M. B., 28

Christie, B., 98–99

Christ, R. E., 821

Chromostereopsis, 821, 821*f*

Chunking, 70*b*, 71–72, 712–713

Churchill, E. F., 14, 103

Classic system-centered error message, 800, 801*f*

Clement, A., 398–399

Click-through wireframe prototypes, 422

Cockton, G., 556, 604, 621

Code testing, 635

Codiscovery, 449, 558*b*

- recruit for, 493, 493*b*
- think-aloud technique, 512

Cognitive affordances, 98*b*, 371*b*, 654, 654*t*, 685, 690*f*, 704*b*, 828–830, 828*f*

- avoiding errors with, 764–765
- clarity of, 733
- cognitive directness in, 758–760
- cognitive support information, 655–656
- completeness of data, 761–762, 762*f*
- considerations, 687
- consistency, 739–745, 740–744*b*, 740–741*f*
- content and meaning, 732–767, 732*f*
- controlling complexity, 746–752, 746–751*f*, 747–752*b*
- defined, 654–655
- design issues, 657–661
- distinguishability of choices, 738–739, 738–739*b*
- for error recovery, 765–766
- existence, 723–725, 723*f*, 725*f*
- human memory limitations, 755–757, 755–757*b*, 756*f*
- legibility, 728
- mail icon as, 656*b*, 656*f*
- for modes, 766–767, 767*b*
- noticeability, 727–728
- physical devices, 665, 665*f*
- precise wording, 733–738, 734–737*b*
- presentation, 725–731, 726*f*
 - complexity, 728
 - consistency, 730–731, 730–732*f*
 - timing, 729–730
- shared conventions, 656–657
- starring role in UX design, 655
- task structure (*see* Task structure)
- useful defaults, 752–755, 752–754*b*
- user choices, 752–755, 752–754*b*
- user/usage centeredness, 762–763
- visibility, 727, 727*f*

- Cognitive complexity theory, 99
- Cognitive directness, 758–760, 758–760*b*, 759*f*
- Cognitive load theory, 714
- Color, 680
- Color-emotion guideline, 822
- Commercial product perspective
 - multi-user collaborative products, 62
 - single-user products, 62
- Common Industry Format (CIF), 592
- Communication, 630
- Competitive analysis, 126–127
- Complexity space, 51, 51*b*
- Computer-based touchscreen voting machines, 251
- Computer graphics, 100
- Computer supported cooperative work (CSCW), 254
- Computer System Usability Questionnaire (CSUQ), 519
- Concept mapping, 40
- Conceptual design, 33, 78*b*
 - as connection between designer and user mental models, 328–340
 - for designing for emotional impacts, 334*b*, 336–337
 - designing the ecology, 331
 - designing the interaction, 331, 336
 - design patterns, 337
 - dimensions of, 340*f*
 - Johnson and Henderson's rule, 334
 - as mapping, 331–333
 - metaphors, 337–339
 - subsystems by work role, 339
 - work practice ecology, 334–335
- Conceptual framework, theory-based, 695
- Constantine, L. L., 76, 193, 204–206, 256–257, 630, 639–640, 823
- Constraints, 282–284
- Consultative design, 399
- Contextual inquiry, 119. *See also* Usage research
 - activity theory, 252–253
 - ethnography, 253–254
 - HCI, 254
 - history, 252–255
 - participatory design, 254–255
- Continuous communication, 630
- Cooper, A., 103, 106, 182, 185, 303, 305, 724
- Cooper, G., 714
- Cooper's in-flight entertainment system, 305*b*
- Cooperative user-system task performance, Interaction
 - Cycle
 - intention shifts, 706–707
 - path variations, 704–705
 - secondary tasks, 706–707
 - stacking, 706–707
- Cordes, R. E., 489
- Costabile, M. F., 12
- Cost-benefit analysis, 24
- Cost-importance analysis, 572*t*, 593–594, 593*b*, 596, 596*b*, 599
 - cost to fix, 575–576
 - cumulative cost, 579
 - defined, 572
 - drawing conclusions, 579–582
 - emotional impact problems, 583–584
 - importance rating adjustments, 573–574
 - importance to fix, 573–574, 574*t*
 - line of affordability, 579, 583
 - multiple problem solutions, 583
 - priority rankings, 578–579
 - priority ratio, 576–578, 577*t*
 - problem groups, 572, 576, 576*t*
 - rapid version, 584
 - resolution, 579–582
 - solutions, 574, 575*t*
 - tie-breakers, 583
- Cost justification, 24–25
- Critical incident technique, 508–509, 610*b*, 614–616, 614*b*, 616*f*
 - data capture
 - avoid video recording, 510
 - defined, 510
 - follow up on hunches, 510
 - manual note taking, 510
 - defined, 508–509
 - identification, 448, 509
 - variation, 509

Critical thinking, 45
 Critiquing, 271*b*, 322–324, 324*f*
 rules of engagement, 324–325
 Cross-disciplinary approach, 10
 Cross, N., 44, 310
 Cultural convention, 657
 Cultural interaction, 12
 Cumulative cost, 579
 Curson, I., 28, 604
 Curtis, B., 28
 Customer service and support, 5, 21
 Custom style guide, interaction design, 381–382

D

Dagstuhl framework, interaction, 21
 Dancing bear software, 106
 Data analysis, 143–145
 cost-importance analysis, 571–584
 domain complexity, 145*b*
 process feedback, 584–585
 qualitative, 565–571
 quantitative, 561–565
 synthesis, 145*b*, 176
 for system, 586–587
 work activity affinity diagram, 153–176
 work activity notes
 additional information, 150
 brief and concise, 147–148
 into categories, 151
 completeness, 148
 context retention, 148–149
 data sources, 150–151
 elemental note, 146–147
 extraction, 151–153
 handwritten/typed, 146
 organization, 154–155
 Database system, 790–792, 791*b*
 Data capture, 139–140
 Data collection method, 501
 empirical UX evaluation, 505–508
 critical incident data capture, 509–510
 critical incident identification, 508–509
 in ecological layer, 507–508, 507*b*
 procedures for, 532–537
 qualitative data (*see* Qualitative UX evaluation)
 quantitative data (*see* Qualitative UX evaluation)
 think-aloud data collection techniques, 511–512
 Data elicitation, usage research
 client company/organization, 124
 collection of, 131–142
 concepts, 119–121
 goals, 121–123, 128
 interviewing *vs.* observing, 131
 preparation for, 124–131
 proposed product/system, 124–125
 subject domain, 124
 team organization, 128–129
 Data loss risk, 57, 177–178
 Data modeling, usage research
 analysis, 179–180
 artifact model, 179*b*
 construction, 208–210
 defined, 207–208
 elicitation, 179–180
 flow model, 188–190*b*
 central importance, 187
 customer journey map, 192
 defined, 187
 making of, 188–190
 for product/system, 190–192
 hybrid model, 221
 information architecture model, 214–215
 kinds of, 178–179
 model consolidation, 221–222
 physical work environment model
 environmental factor, 212–214
 hardware design, 212
 social model
 active entity, 216–217
 concerns, 219–220
 influences, 219–220
 issues with, 217–219
 for product/system, 220
 shared workplace culture, 216

- simplified approach, 216
- smartphone usage, 220
- suggestions for, 179–180
- tagging model with source ID, 180
- task sequence model, 200*b*, 206
 - beyond linear model, 203
 - components, 199–200
 - defined, 197–198
 - essential use case, 204–206
 - menu planning application, 197–198*b*
 - for product/system, 198–199
 - state diagrams, 206
 - step-by-step description, 200–202
 - task interaction model, 201–202*b*
- task structure model (*see* Hierarchical task inventory (HTI))
- user personas
 - creation, 186
 - data extraction, 185–186
 - defined, 185
 - early sketch, 187
 - Ticket Kiosk System, 186*b*
- work role model, 181–182*b*
 - defined, 181–182
 - mediated work role, 182
 - for product/system, 182–183
 - subroles, 182
 - user class, 183–184
- wrap up
 - barrier summaries across all model, 222–223
 - UX design studio, 223–225
- Data note, elemental, 180, 180*b*, 567, 567*b*
- Data organization, 40–41
 - card sorting, 155
- Dayton, D., 592
- Dayton, T., 831
- Dearden, A. M., 131
- del Galdo, E. M., 508–509
- Design, 44*b*, 265*f*, 299*b*, 302*b*
 - creation sub lifecycle, 271–277
 - defined, 264–266
 - guidelines (*see* User experience (UX) design guidelines)
 - human needs, satisfying, 267–268
 - interdisciplinary nature, 264
 - iteration
 - conceptual, 273
 - detailed, 275–276
 - generative, 271–272
 - intermediate, 274–275
 - refinement, 276
 - lifecycle process, 29–30, 30*f*, 34
 - mental models, 401
 - metaphor, 266*b*
 - as noun, 265–266
 - participatory design, 397*b*
 - history, 398–399
 - origins, 398–399
 - PICTIVE, 399–400
 - principles, 710
 - production, 34, 377–380
 - purpose of, 267–268
 - refinement iteration, 276
 - reviews, 547–549
 - rights, 35–36
 - scenario, 197–198
 - SE implementation, 276
 - situations, 47
 - synthesis, 295, 295*b*
 - universality, 263
 - as verb, 266
 - walk-throughs, 547–549
- Design creation. *See* Generative design
- Designers, 295
 - mental model, 328, 329*f*, 330, 336
 - workflow, 335*f*
- Design patterns, 311, 362, 382*b*
 - in conceptual design, 337
- Design solutions lifecycle activity, 262*f*, 267*f*, 294*f*, 342*f*, 360*f*, 386*f*
 - macro view, 272*f*
- Design studio. *See* UX design studio
- Design techniques as life skills, UX

Design techniques as life skills, UX (*Continued*)

- abstraction, 39–40
- brainstorming, 43
- critical thinking, 45
- data/idea organization, 40–41
- framing and reframing, 44
- immersion, 42
- iteration, 45
- modeling, 41
- note taking, 40
- observation, 37–39
- prototyping and envisioning, 45
- reasoning and deduction, 44–45
- sketching and drawing, 43
- storytelling, 41–42

Desmet, P., 525–526

Detailed design iteration, 274–275, 275*f*

Dick, W., 438

Diehl, V. A., 515

Digital Equipment Corporation, 254

Direct interaction, 6

Discount UX evaluation method, 548*b*

- be practical, 605
- criticism, 604
- defined, 603
- limitations, 604
- rigorous methods, 604

Discovery and Analysis Resource (DARe) model, 556

Domain complexity, 51–52, 52*f*, 145*b*

Donohue, J., 297

Dormann, C., 522, 617

Dorst, K., 44, 310

Double experts, 491

Dourish, P., 105, 312

Draper, S. W., 281, 449, 512, 659–661

Dray, S., 450

Drill-in concept, 274

Dual experts, 126

Dubberly, H., 12, 16, 66, 103

Dumas, J. S., 568, 591, 598

Dygdon, J. E., 138

Dzida, W., 615

E

Eardley, R., 104

Ecological approach, 655

Ecological design, 270*b*, 341–343

- conceptual design, 349–351*f*
- creation of, 346–350
- interaction channel, 344
- multiple interaction channels, 344–345
- smart shopping application, 352–358, 355*f*
- usage context, 343
- user behavior, 350–352
- user frustration, 345–346*b*

Ecological-layer inspection, 559

Ecological perspectives, 268, 269*b*, 348*b*, 372*b*

Ecological validity, 468*b*, 470–471

E-commerce software systems, 626

Ede, M., 604

Efficient task path, planning, 720

Ehn, P., 252, 398–399, 407

Ekman, P., 617

Elemental data note, 146–147, 180, 180*b*, 567, 567*b*

- user persona model, 186
- workflow model, 180

Email system, 766, 800, 800*f*

Embodied interaction, 102–105, 312–313, 312–313*b*, 355*b*, 427*b*, 618*b*

Emotional affordances, 654*t*, 685, 829*b*

- considerations, 689
- defined, 684
- meaningfulness, 685
- physical devices, 665

Emotional impact design, 8, 9*f*, 10–15, 126*b*, 244, 244*b*, 385–387, 427*b*, 459*b*, 485*b*, 514*b*

- conceptual design, 389–390, 395
- creation of, 387–395
- distinctiveness, 387
- evaluation questionnaires, 520
- intermediate design
 - audio characteristics, 394
 - motion styles, 392–393
 - physics of interaction, 392–393
 - social and psychological aspects, 394

- tone of language, 393–394
- visual language and vocabulary, 391–392
- market differentiator, 387
- production, 394–395
- pyramid of user needs, 385*b*
- start with inputs, 388
- Emotional-layer inspection, 559–560
- Emotional perspectives, 268, 385*b*
- Emotional satisfaction, 11
- Empirical UX evaluation method, 441, 441*b*
 - baseline level, 476
 - benchmark tasks
 - crafting, 465–468
 - defined, 460
 - mechanics, 469–471
 - selection of, 460–465
 - data collection method, 505–508, 541–543
 - critical incident, 508–510
 - in ecological layer, 507–508, 507*b*
 - procedures for, 532–537
 - qualitative data (*see* Qualitative UX evaluation)
 - quantitative data (*see* Quantitative UX evaluation)
 - think-aloud data collection techniques, 511–512
 - data collection sessions
 - interaction with participants, 535–536
 - session begins, 535
 - wrapping up, 536–537
 - goals, 457–458, 486–487, 507
 - lab-based evaluation, 442–443
 - measures, 458–459
 - metrics, 474–476
 - observed results, 480
 - pilot testing, 503
 - preparation for session
 - data collection forms, 501
 - full lifecycle iterations, 497
 - informed consent, 497–500
 - lab and equipment, 494–496
 - nondisclosure agreements, 501
 - paperwork, 500–501
 - parameters, 496–497
 - questionnaires and surveys, 501
 - task and session lengths, 496–497
 - training materials, 502
 - project context, 454–455
 - quasiempirical, 443
 - rapid approach, 481
 - recruiting participants
 - budget establishment, 490
 - for codiscovery, 493
 - database, 492
 - expert participants, 490–491
 - hard-to-find participant types, 492–493
 - identification of participants, 490–491
 - incentives and remuneration, 492
 - management, 493
 - recruiting methods and screening, 491–492
 - subsequent iterations, 493–494
- RITE, 443
- scope and rigor, 485–486
- select team roles
 - facilitator, 487
 - participation and buy-in, 487
 - prototype executor, 487
 - qualitative data collectors, 488
 - quantitative data collectors, 487
 - supporting actors, 488
- setting levels, 476–480
- target level, 476
- target tables, 455–456, 481
- user classes, 456
- user satisfaction questionnaires, 472–473
- user tasks
 - benchmark tasks, 488
 - exploratory free use, 489
 - unmeasured tasks, 488–489
 - user-defined tasks, 489
- Wheel lifecycle process, 454*f*, 484*f*, 506*f*
- work package, 502
- work role, 456
- Engel, S. E., 831
- Engineering paradigm, 107–108
- Engineering UX evaluation, 439–441, 440*f*

Enterprise system, 62, 88*b*, 119*b*, 123, 127
 Ergonomics, 96–97
 Error proofing, 764–765
 Essential use case task sequence models, 204–206
 Ethnography, 122, 253–254
 Evaluation, UX lifecycle activity, 36–37
 Evaluator effect, 603, 612–613
 Evenson, S., 66
 Experience prototyping, 430–431
 Expert evaluation, 551
 Expert participants, 490–492
 Expert UX inspection, 445, 551
 eXtreme programming (XP), 627

F

FaceAPI system, 617–618
 Face-to-face communication, 599
 Faiola, A., 429
 False cognitive affordances, 659–661
 in door sign, 659*b*, 659*f*
 in microwave control, 660–661*b*
 in webpage links, 660*b*
 Fano, A., 102
 Faris, J., 831
 Feedback
 clarity of, 796–797
 completeness, 797–799
 consistency, 801–804
 content and meaning, 795–804, 795*f*
 existence of, 790–792, 791*f*
 expression, 799
 precise wording, 797
 usage centeredness, 799–801
 user control over feedback detail, 804
 Feedback presentation, 792–795, 792*f*
 complexity, 794
 consistency, 794
 legibility, 793
 medium, 794–795
 noticeability, 793
 timing, 794
 visibility, 793
 Feed-forward cognitive affordances, 724
 Feiner, S. K., 100–101
 Fiegel, T., 573
 Fitts, P. M., 97, 99, 614, 667, 669
 Fitts' law, 669, 670*t*, 778, 781
 Flanagan, J. C., 614
 Flanagan's critical incident techniques, 614
 Flat design style, 10
 Flatt, J. D., 102
 Flow model, 169–170, 178–179, 188–190*b*, 288*b*, 305–306, 306*b*
 central importance, 187
 customer journey map, 192
 defined, 187
 making of, 188–190
 non-UI software functionality, 188
 for product/system, 190–192
 Focus group, 126, 126*b*, 549
 Foley, J. D., 100–101
 Forlizzi, J., 22, 104
 Formal summative UX evaluation, 437*b*, 438–439, 440*t*, 490*b*, 592*b*, 602*b*
 sampling, 608
 Formative testing, 592
 Framing, 42*b*, 44
 Frank, B., 183
 Frederick Winslow Taylor scientific management, 96
 Friction, 814–815
 Friesen, W., 617
 Frishberg, N., 427–428
 Frøkjær, E., 590
 Fulton, B., 443, 537–538
 Functional affordances, 654*t*, 683*f*, 685, 830–831
 considerations, 687
 defined, 683–684
 Fun interaction, at work, 109–110
 Funnel model, 625*b*
 agile UX, 75*f*, 625, 625*b*
 early activities, 78–80
 late activities, 76–77
 need for, 74–75
 scope in, 75

- speed and rigor, 75–76
- speed kills, 74
- low rigor/high speed *vs.* high rigor/slow speed, 76*f*

G

- Gajendar, U., 290
- Gallup, R., 554–555, 558
- Gardiner, M. M., 98–99
- Gardner, P. H., 523, 618–619
- Gaver, W. W., 652, 826–827
- Gellersen, H., 103
- General-purpose usability questionnaires, 519–520
- Generative design, 33, 361
 - activities, 296
 - critiquing, 296*f*, 322–324
 - ideation, 296*f*
 - aircraft flight recorders, 302
 - building blocks, 297–298
 - catalysts, 298, 308–310
 - creative role, 297
 - informers, 298, 302–305
 - practice in, 320–322
 - scope, 298
 - techniques, 298, 310–313
 - immersion, 295
 - iteration, 271–272, 271*b*, 273*f*
 - low-fidelity prototypes, 417–418
 - preparing for, 295
 - rules of engagement, 324–325
 - sketching, 296*f*
 - characteristics, 314–316
 - physical mockups, 322
 - practice in, 320–322
 - synthesis, 295, 295*b*
- George, J. F., 397–398
- George, P., 606
- Gershman, A., 102
- Gibson, J. J., 655, 665, 826
- Gibson's ecological view, 655, 665*b*, 684, 830–831, 830*b*
- Giesecke, F. E., 138
- Gilb, T., 606
- Gillham, R., 53–54
- Gladwell, M., 308
- Glina, V., 103–104, 313–314
- Goals, Operators, Methods, and Selections (GOMS)
 - family of models, 99
- Goble, J. C., 101
- Good, M. D., 254, 431, 606
- Gothelf, J., 232
- Granda, R. E., 831
- Graphical user interface (GUI), 87, 667, 709, 715, 765, 831
- Graphic design, 425
- Graves, M., 43
- Gray, W. D., 442, 604, 695
- Greenbaum, J. M., 399
- Green, T. R. G., 99
- Gross, M., 105, 312–313
- Group affinity labels, 161*f*
- Grudin, J., 100, 810–811
- GUI. *See* Graphical user interface (GUI)
- Gulliksen, J., 253
- Gundelsweiler, F., 548, 635, 640, 645
- Gunn, D. V., 617

H

- Hackman, G., 512
- Hafner, K., 15–16
- Hair dryer control switch, physical overshoot, 782, 782*f*
- Hall, L., 556
- Hallvard, T., 362
- Hammer, A. L., 493
- Hammond, N., 98–99
- Hamner, E., 104
- Hanington, B. M., 37, 155, 549
- Hansen, W., 730–732
- Haptics, 672*b*, 783–784
- Haptic sensory issues, 682–683
- Harper, R., 104
- Harrison, S., 46–47, 99, 107–109, 162–163
- Harslem, E., 101
- Hartson, R., 106, 408, 441–442, 444, 450, 508–509, 615, 651, 827–828
- Hassenzahl, M., 10–11, 109–110, 524–525
- Hazzan, O., 39

- HCI. *See* Human-computer interaction (HCI)
- Health organization, 22
- Hedonic phase, 108
- Held, T., 525
- Heller, F., 428
- Henderson, A., 330, 333–334
- Hertzum, M., 603, 612–613
- Heuristic-based expert UX inspections, 554
- Heuristic evaluation (HE), 445, 445*b*, 551, 555*t*, 556–557
 documenting UX problems, 554
 heuristics, 552–553
 limitations, 556–557
 procedure, 553–554
 variations, 554–556
- Heuristics, 552–553
- Hewett, T. T., 832
- Hierarchical task inventory (HTI), 180*b*, 236, 236*b*, 366*b*
 advantages, 193
 creation, 193
 first-level diagram, 195*b*, 195*f*
 hierarchical relationship, 193–194, 194*f*
 for product/system, 195–197
 task structure model, 192
 benefits, 192–193
 vs. functions, 193
 temporal sequencing, 194–195, 194*f*
 user work role, 195, 196*f*
- High-fidelity physical prototyping, 430–431
- High-fidelity wireframes, 378–379*f*, 415*f*, 420*f*
- Hill, H. C., 138
- Hinckley, K., 101
- HIP. *See* Human information processing (HIP)
- Hix, D., 106, 821
- Hollan, J. D., 101, 697
- Hollands, J. G., 269
- Holtzblatt, K., 14, 28, 187, 199, 231–232, 254, 262, 281–282, 455–456, 606, 633, 635–636
- Honan, M., 810
- Hoonhout, J., 5
- Horizontal prototype, 408, 408*f*, 423*b*
- Hornbæk, K., 590
- Houben, S., 344
- Ho, W., 826–827
- Howarth, D., 692*f*
- HTI. *See* Hierarchical task inventory (HTI)
- Hudson, J. M., 14
- Hudson, S. E., 101
- Hudson, W., 609
- Hughes, D. E., 682
- Hughes, J. A. F., 100–101, 254
- Human-artifact interaction, 4*f*
- Human-centered design (HCD), 281–282
- Human-centered interaction, 827–828
- Human-computer interaction (HCI), 5, 46, 652, 669, 831–832
 changing concepts
 disappearing technology, 102
 embedded, ubiquitous, and ambient interaction, 102–104
 situated, embodied, and tangible interaction, 104–105
 contextual inquiry, 254
 early industrial engineering, 96–98
 Frederick Winslow Taylor scientific management, 96
 hardware and software foundations, 99–101
 history, 95–106
 human factors, 96–99, 96*b*
 participatory design, 397
 shifting paradigms
 design and development, 109
 engineering paradigm, 107–108
 human information processing paradigm, 108
 phenomenological paradigm, 108–109
 usability, 9–10
 WW II, 98
- Human factors engineering, 96–99, 107*b*
- Human information processing (HIP), 98–99, 99*b*, 107–108
- Human-Machine Interaction Network on Emotions project, 618–619
- Human memory
 chunking, 712–713
 in cognitive affordances, 755–757, 755–757*b*, 756*f*

- cognitive load, 714
- kinds of memory, 715–716
- long-term memory, 716
- muscle memory, 716
- phone number design, 713*b*
- recognition *vs.* recall, 714–715
- sensory memory, 715
- shortcuts, 715
- short-term/working memory, 712–715
- stacking, 713–714
- Hutchins, E. L., 101, 697
- Hybrid model, 221
- I**
- Iannella, R., 831
- Ideal mental model, 328, 330
- Ideation, 105*b*, 296*f*, 299*b*, 318*f*
 - aircraft flight recorders, 302
 - critiquing, 300*b*
 - framing, 310, 310*b*
 - generative design
 - aircraft flight recorders, 302
 - building blocks, 297–298
 - catalysts, 298, 308–310
 - creative role, 297
 - informers, 298, 302–305
 - practice in, 320–322
 - scope, 298
 - techniques, 298, 310–313
 - at IDEO, 299–300*b*
 - Magic wand technique, 310–311
 - physical mockup, 300*b*
 - reframing, 310
 - rules of engagement, 324–325
 - sketching, 315
- Immersion, 42, 84*b*, 135*b*, 176*b*, 187*b*
- Implementation coding, 634
- Incubation ideation technique, 297*b*, 311
- Indirect interaction, 6
- Individual detection rate, 613
- Informal communication, 630
- Informal quantitative data analysis, 562
- Informal summative UX evaluation, 439, 440*t*, 561, 561*b*, 591, 591*b*, 602
- Information, 108, 136, 150, 214–215, 269, 655–656, 804–807
 - architecture model, 136, 136*b*, 170–171, 179, 214–215, 214*b*, 269–270, 307, 308*b*, 363–364
 - design, 270–271
 - display assessment
 - information organization, 804–806
 - visual bandwidth, 807
 - science, 269
- Information architecture, 214–215, 215*b*, 363–364
 - pervasive, 192*b*, 270, 343, 347*b*, 352*b*
- Informed consent, 532*b*
 - form, 498–500, 499–500*b*
 - permission application, 497–498
- Informers, ideation, 298, 299*b*, 302–305
- In situ user research, 138
- Inspection (UX), 37*b*
- Institutional review board (IRB), 497–498
- Intention shifts, 706–707
- Interaction, 18*b*
 - emotional impact, 11–12
 - interrupt-driven, 4
 - user experience
 - activity-based, 19
 - Dagstuhl framework, 21
 - direct/indirect, 6
 - expanding concept, 3–4
 - localized, 18
 - system-spanning, 20–21
- Interaction artifact model, 686*b*
 - defined, 3
 - human–artifact interaction, 4*f*
- Interaction channel, 682*b*
- Interaction complexity, 51, 52*f*
- Interaction Cycle
 - affordances, 825–827
 - cognitive, 828–830
 - confusion over HCI/UX, 827–828
 - functional, 830–831
 - Gibson’s ecological view, 830–831
 - assessment

Interaction Cycle (*Continued*)

- content and meaning of feedback, 795–804
- existence of feedback, 790–792
- information displays, 804–807
- presentation of feedback, 792–795
- system feedback, 789–790
- automation issues, 786–789
- business report creation, 703–704
- defined, 695
- Norman's stages-of-action model, 699
 - gulfs between user and system, 697–698
- outcomes, 784–789, 785*f*
- overall guidelines
 - accommodation of user differences, 823
 - anthropomorphism, 815–818
 - consistency, 810–814
 - helpful help, 823–824
 - humor, 815
 - reducing friction, 814–815
 - simplicity, 808–810
 - text legibility, 822
 - tone and psychological impact, 818
 - use of sound and color, 818–822
 - user preferences, 822
- physical actions
 - sensing objects, 778–779
 - user help, 779–783
- planning, 717–722
- schematic diagram, 717*f*
- structure, 717, 717*f*
- system functionality, 785
- system response time, 785
- theory-based conceptual framework, 695
- translation, 722–777
- user planning
 - efficient task paths, 720
 - progress indicators, 720–721
 - system task model, 718–720
 - transaction completion slips, 721–722
- user-system task performance
 - intention shifts, 706–707
 - path variations, 704–705

- primary tasks, 704
- secondary tasks, 706–707
- stacking, 706–707

UX design issues

- assessment, 702–704
- outcomes, 702
- physical actions, 701–702
- planning, 700
- translation, 701

Interaction design, 380*f*

- conceptual design, 361–362
- creation of, 361–365
- custom style guide
 - defined, 381–382
 - usage of, 382
- defined, 359–360
- device types, 360–361
- ecology roles, 361
- information architecture, 363–364
- patterns and techniques, 362
- production, 377–380
- storyboards
 - between-frame transitions, 369–371
 - defined, 365–366
 - pyramid of user needs, 366–369
- supporting tasks, 359–360
- wireframes
 - defined, 373–374
 - path to, 372

Interaction-layer inspection, 559

Interaction technique, 101, 268, 322*b*, 351*b*, 364*b*

- conceptual design, 365*f*

Interactive graphics, 100

Intermediate design, 33

- emotional impact design
 - audio characteristics, 394
 - motion styles, 392–393
 - physics of interaction, 392–393
 - social and psychological aspects, 394
 - tone of language, 393–394
 - visual language and vocabulary, 391–392
- iteration, 274–275, 274*f*

Internet, 6
 Interrupt-driven interaction, 4
 Irby, C., 101
 Isaacson, W., 810
 Ishii, H., 105, 312–313, 344
 Israr, A., 652
 Iteration design, 45, 92
 conceptual, 273
 detailed, 275–276
 generative, 271–272
 intermediate, 274–275
 refinement, 276
 iTunes ecology, 19*b*, 342–343*b*
 Ivanov, Y., 619
 Izadi, S., 104

J

Jacob, R. J. K., 101
 Jacobsen, N. E., 603
 Janda, A., 96
 Jeffries, R., 591, 598
 JIRA software, 633
 John Hancock problem, 786–788*b*, 787*f*
 Johnson, D. C., 102
 Johnson, J., 330, 333–334, 733
 Joint Application Design, 399
 Jokela, T., 710
 Jones, B. D., 12
 Jones, P., 42
 Jones, R. E., 97, 614
 Jones, S. J., 431

K

Kachur, O., 42
 Kammersgaard, J., 398–399
 Kane, D., 155
 Kantrovich, L., 814
 Kapoor, A., 619
 Karat, C. -M., 573
 Karyukina, B., 604
 Kassell, N. F., 101
 Kaur, K., 706

Kawakita, J., 155–156
 Kellogg, W. A., 253, 338, 470
 Kennedy, S., 449, 512
 Kennedy, T. C. S., 831
 Kensing, F., 131, 408
 Kern, D., 683
 Keystroke level model, 99
 Kieras, D. E., 99
 Kimball, R., 101
 Kim, J. H., 520, 617
 Kindlund, E., 604
 King, V., 254
 Kirakowski, J., 604
 Kjaer, A., 529
 Klokrose, C., 344
 Knemeyer, 7–8
 Knox, S., 254
 Koenemann-Belliveau, J., 615
 Kolko, J., 7–8, 16, 229
 Koller, F., 524
 Kortum, P. T., 521, 563
 Kramer, J., 39
 Krippendorff, K., 308
 Krueger, R. A., 126
 Kugler, L., 682
 Kuhn, S., 255, 398
 Kuniavsky, M., 103
 Kyng, M., 252, 398–399, 407

L

Lab-based UX evaluation, 436*b*, 442–443, 615
 Lafrenière, D., 255
 Lamb, M. R., 101
 Landauer, T. K., 551, 609, 612, 830–831
 Landry, S., 90
 Lang, P. J., 526
 Lantz, A., 253
 Lanzilotti, R., 12
 Large-scope approach, 60
 Last-in-first-out (LIFO) data structure, 713
 Lathan, C., 104
 Laugwitz, B., 525

- Lavery, D., 556, 621
 - Lavie, T., 520
 - Law, E. L., 5
 - Learnability, 574
 - LeCompte, M. D., 253
 - Lee, C., 104
 - Lee, J., 12
 - Legacy system, 124*b*, 245*b*
 - Lehner, K., 525
 - Lewis, C., 511, 696–697
 - Lewis, J. R., 519, 592, 609, 609*f*
 - Lewis, R. O., 68
 - Lewis, S., 254
 - Li, N., 11–12
 - Likert, R., 515
 - Lincoln, J. E., 831
 - Liu, M., 11, 515, 522
 - Lickorish, A., 769
 - Lifecycle activities
 - concepts, 33
 - design solutions, 33–36
 - evaluation, 36–37
 - mapping project parameters, 47, 48*f*
 - prototype candidates, 36
 - understand needs, 33, 34*f*
 - UX design lifecycle process, 30*f*
 - Lifecycle process approach, agile SE
 - planning
 - controlling scope, 633
 - customer stories, 631–633
 - development tasks, 633
 - story-based, 632–633
 - sprints
 - acceptance test creation, 634
 - acceptance testing and deployment, 635
 - code testing, 635
 - implementation coding, 634
 - regression testing, 635
 - unit code test creation, 634
 - Lifecycle process choice, UX, 46
 - Lifecycle sub-activities, 30–31
 - Localized interaction, 18, 21
 - Local prototypes, 409
 - Lockwood, L. A. D., 193, 204–206, 256, 639–640
 - Long-term interaction, 21
 - Long-term memory, 716
 - Lotter, M., 104
 - Low-fidelity prototypes, 417–418, 419*f*
 - Low-fidelity sketched wireframes, 419*b*
 - Löwgren, J., 430
 - Low-rigor approach, 50
 - Lubow, A., 291
 - Lund, A. M., 518, 831
 - Lynch, G., 254
- ## M
- MacKenzie, I. S., 99, 669
 - Mack, R. L., 253, 338
 - Macleod, M., 28
 - Macredie, R. D., 527
 - Madsen, K. H., 529
 - Magic wand ideation technique, 310–311
 - Maiden, N., 706
 - Manage finances, 195
 - Mantei, M. M., 24
 - Manual dexterity, 669, 781
 - Manual note taking, data collection, 510
 - Marketing survey, 120
 - Markopoulos, P., 104
 - Marquardt, N., 344
 - Marsh, George Perkins, 38–39
 - Martin, B., 37, 155, 549
 - Mason, J. G., 324–325
 - Mateas, M., 254
 - Matheson, L., 554–555, 558
 - Maurer, F., 638–639
 - Mayhew, D. J., 24, 106
 - McCaulley, M. H., 493
 - McCrickard, S., 162–163
 - McCullough, M., 105
 - McGee, M., 443, 537–538
 - McGrenere, J., 826–827, 830–831
 - McInerney, P., 638–639
 - Meads, J., 256, 639, 641

- Meaningfulness, 8, 9*f*, 14*b*, 16, 388*f*, 521, 685, 685*b*
 - generating and collecting, 526–532
 - goals of, 528
 - long-term studies, 527–528
 - periodic questionnaires, 530
 - consistency rules, 596–597
 - reporting vocabulary
 - jargon, 597
 - precision and specificity, 597
 - report tone
 - large amounts of qualitative data, 598–599
 - personal presence, 599
 - respect feelings, 598
- Mediated users, 182
- Medium-fidelity wireframes, 422
- Medium- to high-fidelity click-prototypes, 422–425
- Medlock, M. C., 443, 537–538
- Melguizo, M. d. C. P., 330
- Memmel, T., 548, 635–636, 640, 645
- Mental models, 401
 - conceptual design
 - as connection, 328–333
 - design patterns, 337
 - emotional impacts, 336–337
 - interaction, 336
 - as mapping, 331–333
 - metaphors, 337–339
 - subsystems by work role, 339
 - work practice ecology, 334–335
 - defined, 327
 - designer's, 329*f*, 330
 - ideal, 330
 - metaphor, 331*b*
 - user's, 330–331
 - Wheel UX lifecycle, 327, 328*f*
- Menu planning application, 197–198*b*
- Metaphor, 266*b*, 301*b*, 307*b*, 347*b*, 361*b*
 - conceptual design, 337–339, 338–339*b*
 - emotional conceptual design, 389
- Microsoft system designers, 788
- Middleburg University Ticket Transaction Service (MUTTS), 156*b*, 181*b*, 234*b*, 245*b*, 347*b*, 477*b*
 - existing system, 88–89
 - organizational context, 89
 - flow model, 189–190*b*, 190–191*f*
 - consolidation, 221–222*b*, 222–224*f*
 - hierarchical task inventory diagram, 195*f*
 - ticket seller role, 196*f*
 - information architecture modeling, 215*b*
 - physical work environment model, 212–214*b*, 213*f*
 - rationale, 90
 - simple early flow model, 188–189*b*, 189*f*
 - social model, 217*b*, 219–220*b*
 - task sequence model
 - branching and looping, 203*b*, 204*f*
 - rudimentary, 200*b*
 - step-by-step task interaction model, 201–202*b*
 - ticket buying, 206
 - user class definition, 184*b*
 - work activity affinity diagram, 156, 163, 164*f*
 - work role identification, 181–182*b*
- Miller, G. A., 712
- Miller, J. T., 521, 563
- Miller, L., 637, 639
- Milroy, R., 769
- Mind-mapping, 40–41
- Minimum viable product (MVP), 71, 232
- Mini-UX check-list, 643, 644*t*
- Mitchell, A., 138
- Mobile applications, 429
- Model consolidation, 221–222
- Modeling, 41
- Moggridge, B., 427
- Molich heuristics, 619–620
- Molich, R., 445, 551–553, 568, 591, 598, 604, 608–609, 619–620
- Mond, N. E., 831
- Mood boards, 274*b*, 295*b*
 - conceptual design, 389–390, 390*f*
 - National Parks Website, 389*b*
- Moon, J. Y., 520
- Moore, N. C., 90, 291

Moran, T. P., 99, 108
 Morville, P., 270
 Mosier, J. N., 831
 Mrazek, D., 591–592
 Muller, M. J., 20, 20*f*, 255, 297, 398–400, 554–555, 558, 831
 Multimedia system evaluation questionnaires, 520
 Multipurpose screwdrivers, 812–813, 812–813*f*
 Multiuser collaborative products, 62
 Multiuser tasks, 705
 Mumford, E., 399
 Munk-Madsen, A., 408
 Murano, P., 816–817
 Murphy, R. R., 104
 Muscle memory, 716
 Myers, B. A., 101
 Myers, I. B., 493
 Myers-Briggs test, 493

N

Nass, C. I., 817–818
 National Institute of Standards & Technology (NIST), 592
 National Park Project, 374*b*
 color scheme, 391*b*
 custom style guide, 383*f*, 383*b*
 iconography, 392*b*, 393*f*
 mood boards, 389*b*
 Natural mapping, 758
 Nayak, N. P., 591–592
 Network services tool, 804, 805*f*
 Newell, A., 99, 108
 Newman, W. M., 100, 604
 Nielsen, J., 8*np*, 106, 408, 408*f*, 445, 511, 551–553, 604, 608–609, 612, 619–620, 831
 Nilsson, P., 314
 Nielsen heuristics, 619–620
 Nondisclosure agreements (NDAs), 501
 Norman, D. A., 10–13, 19–20, 101, 139, 281, 330–331, 333–334, 345–346, 348, 514, 652, 654, 656, 668, 691, 693, 695, 696*f*, 697, 704–705, 710–711, 758, 769–770, 785, 808–809, 825, 827
 Norman, K. L., 515
 Norman's stages-of-action model, 696*f*
 analytic evaluation method, 696–697
 cognitive walkthrough, 696–697
 Interaction Cycle
 gulf of evaluation, 698
 gulf of execution, 697–698
 gulfs between user and system, 697–698
 outcomes and a system response, 699–700
 partitioning the model, 699
 transition, 699*f*
 Note taking, 37, 40
 Novak, J. T., 138
 Nowell, L., 821

O

Objective UX evaluation data, 437*b*, 458, 458*b*, 518*b*
 Obrist, M., 652
 Observation, 37–39
 of car wash workflow, 38*f*, 38–39*b*
 Olsen, D. R., Jr., 101
 Olson, J. R., 330–331
 O'Malley, C., 449, 512
 Onion-layers effect, 585
 Ottersten, I., 314
 Outcomes, stage of Interaction Cycle, 702
 Overall guidelines, UX design
 accommodation of user differences, 823
 anthropomorphism, 815–818
 consistency, 812–814
 against innovation, 814
 structural consistency, 811–812
 helpful help, 823–824
 humor, 815
 reducing friction, 814–815
 simplicity, 808–810
 text legibility, 822
 tone and psychological impact, 818
 use of sound and color, 818–822
 user preferences, 822

P

Page, C., 554–555, 558
 Pagulayan, R. j., 617

- Palen, L., 531
- Palmiter, S., 254
- Pangaro, P., 12
- Paper-in-device mockup prototypes, 429
- Paradigms, of user experience, 101*b*
- Participants, 120*b*, 439*b*, 448*b*, 460*b*, 485*b*, 508*b*, 546*b*, 563*b*
 - anonymity, 590–591
 - attitude toward UX problems, 533
 - detection rates, 610–611
 - low-fidelity prototypes, 534–535
 - paperwork, 533
 - partnership cultivation, 533–534
 - preliminaries with, 532–533
 - session protocol, 533–534
 - think-aloud techniques, 511–512
- Participants recruitment
 - empirical UX evaluation session
 - budget establishment, 490
 - for codiscovery, 493
 - database, 492
 - expert participants, 490–491
 - hard-to-find participant types, 492–493
 - identification of participants, 490–491
 - incentives and remuneration, 492
 - management, 493
 - recruiting methods and screening, 491–492
 - subsequent iterations, 493–494
 - training materials, 502
- Participatory design, 72*b*, 254–255, 297*b*, 397*b*
 - heuristic evaluation, 554–555
 - history, 398–399
 - origins, 398–399
 - PICTIVE, 399–400
- Patel, N. S., 682
- Path variations, in interaction cycle, 704–705, 705*f*
- Patton, J., 240, 638–639, 643
- Paulk, M. C., 28
- Pausch, R., 101
- Payoff methods, 442
- Peanut butter theory, 17
- Pérez-Quñones, M. A., 344, 810–811
- Pering, C., 426–427
- Persona model, 88*b*, 303–305, 304*f*. *See also* User persona model
- Perspective-based usability inspection, 555–556
- Pervasive information architecture, 192*b*, 270, 343, 347*b*, 352*b*
- Petersen, M. G., 529, 531
- Pew, R. N., 831
- Pfleging, B., 683
- Phenomenological paradigm, 108–109
- Phillips, B. C., 617
- Photo ops, 136
- Physical action guidelines, UX design
 - to avoid physical overshoot errors, 782–783
 - awkwardness, 779–780
 - Fitts' law, 781
 - haptics, 783–784
 - Interaction Cycle, 701–702
 - manual dexterity, 781
 - overshoot, 670–672
 - physical disabilities, 779–780
 - physicality, 783–784
 - sensing objects
 - to manipulate, 778–779
 - during manipulation, 779
- Physical affordances, 654*t*, 685, 702*b*, 760*b*, 826*b*
 - Add to Cart button, 662*b*, 662*f*
 - considerations, 688
 - defined, 661–662
 - design issues
 - awkwardness, 667
 - disabilities, 667
 - Fitts' law, 669
 - manual dexterity, 669
 - physicality, 668
 - physical overshoot, 670–671
 - user interface objects, 667
 - for opening doors, 664–665, 664*f*
 - starring role in UX design, 662, 664*f*
- Physical awkwardness, 667, 779–780
- Physical disabilities, 779–780
- Physical interactivity, 426–429

- Physicality, 313*b*, 338*b*, 668, 668–669*f*, 668*b*, 779*b*, 783–784, 784*f*
- Physical mockup, 316*b*, 428*f*
 - as embodied sketches, 322, 322*f*
 - physical interactivity, 426–429
- Physical models, 305–306
- Physical prototyping, 427–428
- Physical work environment model, 136, 170, 179, 211*b*, 306*b*
 - environmental factor, 212–214
 - hardware design, 212
- Picard, R. W., 619
- PICTIVE approach, 399–400
- Plaisant, C., 806–807, 815
- Planning guidelines, UX design, 718*f*
 - efficient task paths, 720
 - Interaction Cycle, 700
 - lifecycle process approach, agile SE
 - controlling scope, 633
 - customer stories, 631–633
 - development tasks, 633
 - story-based, 632–633
 - progress indicators, 720–721
 - system task model, 718–720
 - transaction completion slips, 721–722
- Planning, stage of Interaction Cycle
 - efficient task paths, 720
 - progress indicators, 720–721
 - system task model, 718–720
 - transaction completion slips, 721–722
- Platz, A., 525
- Pogue, D., 14, 814
- Poisson process, 609
- Polson, P. G., 99, 696–697
- Post-Study System Usability Questionnaire (PSSUQ), 519
- Potosnak, K., 711
- Practical programming techniques, 100
- Preissle, J., 253
- Presentation timing, paper towel dispenser message, 680–681, 680–681*b*, 680*f*
- Pressman, R., 256–257
- Primary task, 704
- Priority ranking, 578–579
- Priority ratio, 576–578
- Privender, S., 104
- Probability function, 609–610
- Product concept statement, 87*b*
 - defined, 91–92
 - process-related exercises, 92–93
 - Ticket Kiosk System, 91*b*
- Product owner, 87–88
- Product perspectives, 119*b*
- Professional UX designer, 799
- Professional *vs.* personal goal, 218
- Programmable user model, 99
- Progress indicator, 720–721
- Project commission, UX team roles
 - analyst, 87
 - designer, 87
 - graphic/visual designer, 87
 - product owner, 87–88
 - usage researcher, 87
- Prototype executor, 487
- Prototyping
 - activity theory, 407*b*
 - advantages, 406–407
 - animated, 429–430
 - decoy user interface objects, 423–424
 - depth and breadth
 - horizontal prototypes, 408
 - local prototypes, 409
 - T prototypes, 409
 - vertical prototypes, 408–409
 - envisioning, 45
 - experience, 430–431
 - fidelity, 410, 416–426
 - generative design
 - low-fidelity prototypes, 417–418
 - high-fidelity physical prototyping, 430–431
 - high-level task context, 417
 - intertwines with, 405
 - paradigm, 407*b*
 - Scandinavian origins, 407
 - specialized variations, 426–432

- state diagrams, 413–414
- universality, 407
- user workflow, 412
- wireflow, 411–412
- wireframe
 - creation of, 414–416
 - decks of, 418
 - defined, 410
 - design elements, 410–411
 - software tools, 432
 - static low-fidelity, 418–419
- Wizard of Oz, 431–432
- Pulido, D., 429
- Pyla, P. S., 103–104, 313–314, 344, 810–811
- Pyramid of user needs, 341*b*

Q

- Qiu, R. G., 104
- Qualitative data
 - collections, 448–449
 - collectors, 488
- Qualitative UX evaluation, 565*b*
 - analysis preparation steps, 565–566
 - critical incident, 508–509
 - data capture, 509–510
 - defined, 508–509
 - identification, 509
 - variation, 509
 - formative analysis, 565
 - problem data management, 571
 - rapid approach, 571
 - rapid iterative testing and evaluation
 - defined, 537–539
 - variations in, 540
 - raw evaluation data, 565, 567–568
 - steps for, 566–571
 - think-aloud data collection techniques
 - codiscovery, 512
 - defined, 511
 - in empirical evaluation, 512
 - management of participants, 511–512
- Quality assurance (QA), 425
- Quality communication, 589–590
- Quality user experience, 564–565
- Quantitative data
 - collections, 447–448
 - collectors, 487
- Quantitative UX evaluation, 562*b*
 - descriptive statistics, 562–563
 - direct observation, 521–522, 528–529
 - emotional impact, 521
 - AttrakDiff questionnaire, 524–526
 - physiological responses, direct detection of, 526
 - self-reporting techniques, 523
 - think-aloud techniques, 522–523
 - verbal self-reporting techniques, 522–526
 - evaluator-triggered reporting, 531–532
 - meaningfulness data, 521
 - generating and collecting, 526–532
 - goals of, 528
 - long-term studies, 527–528
 - periodic questionnaires, 530
 - objective quantitative data
 - counting user errors, 513
 - timing task performance, 513
 - project management decision, 564–565
 - self-reporting, 529–530
 - diary-based, 530
 - simulated real usage situations, 528–529
 - subjective quantitative data collection
 - modifying questionnaires, 520
 - other questionnaires, 519–520
 - questionnaire for user interface satisfaction, 515–516, 521
 - questionnaires, 514–515
 - semantic differential scales, 515
 - system usability scale, 517–518, 521
 - usefulness, satisfaction, and ease of use, 518–519
 - subjective quantitative questionnaire data, 563
 - summative evaluation method, 591
 - voicemail, 531
- Quasiempirical UX evaluation, 443, 443*b*, 447*b*, 540–541
- Quenk, N. L., 493
- Quesenbery, W., 592, 594, 832

Questionnaire for User Interface Satisfaction (QUIS),
514–516, 516–517*t*
Questionnaires, 501

R

Radio frequency identification (RFID), 103
Radoll, P., 12
Ramey, J., 254
Ranasinghe, N., 652
Rapid iterative test and evaluation (RITE), 443, 447*b*,
537–540
 defined, 537–539
 variations in, 540
Reasoning and deduction, 44–45
Recognition *vs.* recall, 714–715
Recruiting participants, 129–130
Reeves, B., 817–818
Refinement, of product concept statement, 92
Reframing, of UX design, 44
Regression testing, 630, 635, 646
Reimann, R., 103
Reisner, P., 811
Reiterer, H., 344, 548, 635–636, 640, 645
Reliability, 29*b*, 444, 602–606, 602*b*
Remote UX evaluation, 450
Resmini, A., 270, 307, 344, 352, 356
Rhee, Y., 12
Rice, J. F., 821
Rieman, J., 696–697
Ries, E., 71
Rigor, UX lifecycle activities
 complexity, 51–56
 defined, 49–50
 development stage, 57–58
 domain complexity, 51–52
 domain familiarity, 56
 interaction complexity, 51
 project resources, 58
 vs. rapidness, 445–447
 risk aversion, 56–57
 system complexity space, 51
 quadrants, 52–56
Riley, M., 449, 512
RITE. *See* Rapid iterative test and evaluation
 (RITE)
Rodden, T., 254
Rogers, W. A., 12
Rogers, Y., 254
Rollins, A. M., 831
Romero, R., 443, 537–538
Root, R. W., 831
Rosati, L., 270, 307, 344, 352, 356
Rosenblum, L. D., 652
Rosenfeld, L., 270
Rosson, M. B., 270, 438, 442
Roto, V., 5, 10
Rouncefield, M., 254
Royce, W. W., 67, 67*f*, 110–111, 628–629
Russell, D. M., 102
Ruyter, B. d., 104

S

Saddler, H., 831
Safos, C., 104
Salzman, M. C., 442, 531, 604, 695
Savio, N., 12
Scandinavian work activity theory, 252, 399
Scapin, D. L., 711
Scenario, 197, 275*b*, 366*b*
Schleicher, D., 42
Schleifer, A., 362
Schmidt, L., 604
Schöbel, M., 524–525
Scholtz, J., 104
Schöning, J., 344
Schrepp, M., 525
Schroeder, W., 604, 609, 613
Schuh, E., 617
Schulman, R., 821
Scientific management, 96
Scope of delivery, 59–61, 119*b*, 128*b*, 151*b*, 229*b*, 232*b*,
277*b*, 298*b*, 485*b*, 506*b*, 630*b*, 640*b*
SCRABBLE Flash Cube game, 313, 314*f*
Scriven, M., 438, 442

- Sears, A., 555
- Secondary tasks, 706–707
- Seiden, J., 232
- Self-Assessment Manikin (SAM), 526
- Self-reporting technique, 521–522, 529–530
 - diary-based, 530
 - emotional impact data collection, 523
- Sellen, A., 104, 315
- Semantic differential scales, 515
- Sengers, P., 99, 107–108
- Sensory affordances, 654, 654*t*, 685
 - auditory issues, 681–682
 - considerations, 688
 - defined, 672
 - haptic issues, 682–683
 - tactile issues, 682–683
 - visual design issues
 - color, 680
 - discernibility, 675–677
 - distinguishability, 677–680
 - noticeability, 673–675
 - presentation timing, 680–681
 - text legibility, 677
 - visibility, 673
- Sensory memory, 715
- Sequence models, 306
- Served users, 182
- Service design, 22
- Service experience, 22–23
- Session protocol, 533–534
- Shadowing user, 138, 192
- Shared cultural conventions, 656–657, 828–830
- Shattuck, L. W., 614
- Shelly, S., 104
- Sherman, D., 101
- Shewmake, D. T., 73
- Shifting paradigms, for HCI & UX
 - design and development, 109
 - engineering paradigm, 107–108
 - human information processing paradigm, 108
 - phenomenological paradigm, 108–109
- Shih, Y. -H. X., 11, 515, 522
- Shneiderman, B., 101, 555–556, 806–807, 815, 831
- Short-term memory, 712–715
- Sidner, C., 104
- Siegel, D. A., 14, 450
- Silos, 111–113, 629, 629*b*
- Simon, H. A., 712
- Simon, T., 99
- Simonsen, J., 131
- Single-user products, 62
- Singley, M. K., 438, 442
- Situated interactions, 104–105
- Sketching, 296*f*
 - drawing, 43
 - embodied cognition, 316
 - freehand gestural, 317*f*
 - generative design
 - characteristics, 314–316
 - physical mockups, 322
 - practice in, 320–322
 - K-YAN, 319–320*f*
 - laptop projector project, 317*b*
 - rules of engagement, 324–325
- Slanty design, 350–351
- Slivka, E., 15
- Small-scope approach, 60
- Small upfront analysis and design, 640–641
- Smart devices, 6
- Smith, D. R. C., 101, 591–592
- Smith, E. C., 408
- Smith, K. C. M., 101, 308
- Smith, S. L., 831
- SnakeLight, 138*b*
- Sniderman, R., 271
- Snyder, C., 592
- Social interaction, 12
- Social model, 179, 307
 - active entity, 216–217
 - concerns, 219–220
 - influences, 219–220
 - issues with, 217–219
 - for product/system, 220
 - shared workplace culture, 216

- Social model (*Continued*)
 - simplified approach, 216
 - smartphone usage, 220
- Social networking, 351*f*
- Social Security Administration, 255–256
- Software drawing application, 781
- Software engineering (SE), 99–100, 113, 229, 397.
 - See also* Agile software engineering (SE)
 - essential use cases, 256–257
 - funnel model, 80
 - waterfall process
 - lifecycle process, 67–68, 67*f*
 - verification and validation (V and V), 68
- Software Usability Measurement Inventory (SUMI), 519
- Soon, W., 710
- Sorrell, J., 8*np*
- Souza, F. D., 711
- Soviet psychological activity theory, 253
- Spence, I., 256
- Spencer, I. L., 138
- Spine, T., 606
- Spool, J., 604, 609, 613
- Sprints, 237*b*, 298*b*, 631*b*, 633*b*, 641*b*
 - lifecycle process approach, agile SE
 - acceptance test creation, 634
 - acceptance testing and deployment, 635
 - code testing, 635
 - implementation coding, 634
 - regression testing, 635
 - unit code test creation, 634
 - mini agile UX lifecycle process, 77*f*
 - UX integration, 641–643, 642*f*
- Stacking, Interaction Cycle, 706–707, 707*f*
- Stages-of-action model, 695, 711
 - gulf of evaluation, 698
 - gulf of execution, 697–698
- Stake, R., 438
- Standard rigorous method, 59
- Starring role in UX design
 - cognitive affordances, 655
 - physical affordances, 662
- State diagram, 414*f*
- Static low-fidelity wireframes, 418–419
- Stetson, J. N., 516
- Steuer, J., 817–818
- Steve, J., 16
- Stock, M., 269
- Stock, W. G., 269
- Story-based planning, 632–633
- Storyboards, 271*b*, 411*b*, 429*b*, 549*b*
 - interaction design
 - between-frame transitions, 369–371
 - defined, 365–366
 - pyramid of user needs, 366–369
 - transition frame, 371*f*
- Storytelling, 41–42
- Streitz, N. A., 102
- Strijland, P., 832
- Structural consistency, 811–812
- Style guide, 394*b*, 711
- Subjective quantitative data collection
 - questionnaires, 514–515, 519–520
 - lab-based sessions, 514
 - modifying, 520
 - user interface satisfaction, 515–516, 521
 - semantic differential scales, 515
 - system usability scale, 517–518, 521
 - usefulness, satisfaction, and ease of use, 518–519
- Subjective UX evaluation data, 329, 442*b*, 460*b*, 472*b*, 503*b*, 514*b*, 523*b*, 546*b*, 563*b*
- Subject-matter expert (SME), 56, 125–126, 128–129, 445*b*, 604, 604*b*
- Suchman, L. A., 254
- Summative UX evaluation, 561*b*, 601*b*
 - engineering *vs.* science, 601–602
 - formal, 438–439, 602*b*
 - formative, 438–441
 - informal, 439, 602
- Sundblad, Y., 398–399
- Suri, J. F., 6, 430, 532
- Surveys, 501
- Sutcliffe, A., 706
- Sutherland, E. J., 523, 618–619
- Sutherland, I. E., 100

- Sweller, J., 714
- Sy, D., 637
- Synthesis, 145*b*, 176, 295, 295*b*
- System complexity space, 51, 51*b*, 52*f*, 55–56, 229*b*, 240*b*, 303*b*
- quadrants, 52–56
- System feedback assessment, 789–790, 790*f*
- System functionality, 785
- System response time, 785, 789
- System-spanning interaction, 20–21, 20*f*, 307*b*
- System task model, 718–720
- get organized, 718*b*, 719*f*
 - library information system, 720*b*
 - master document feature, 719*b*
- System Usability Scale (SUS), 514, 517–518, 521
- ## T
- Tactile sensory issues, 682–683
- Tangible interaction, 104–105, 312–313, 312–313*b*, 427*b*
- Task analysis, 120
- Task-based expert UX inspections, 554
- Task flow, 462–463
- Task interaction model, 200
- Task model, 170, 192. *See also specific task model*
- Task sequence model, 179, 200*b*, 206, 366*b*
- beyond linear model, 203
 - branching, 203, 203*f*
 - components
 - step goals, 199
 - step trigger, 199
 - task barrier, 199
 - user information and other needs, 199–200
 - defined, 197–198
 - essential use case, 204–206
 - looping, 203, 203*f*
 - menu planning application, 197–198*b*
 - for product/system, 198–199
 - state diagrams, 206
 - step-by-step description, 200–202, 203*f*, 206*b*
 - task interaction model, 201–202*b*
 - usage scenario, 197–198*b*
- Task structure model, 179
- benefits, 192–193
 - for efficiency, 768–769
 - grouping, 769–770
 - for flexibility, 768–769
 - human working memory, 768
 - task thread continuity, 771–774
 - user control, 775–777
 - user work, 774–775
- Tasks *vs.* functions, 193
- Tatar, D., 46–47, 99, 107–108
- Tauber, E. R., 817–818
- Taylor, F. W., 96
- Taylorism, 96
- Teorey, T. J., 24
- Terrano, M., 443, 537–538
- Text legibility, 821*f*, 822
- Thayer, T. A., 110–111
- Theofanos, M., 592, 594
- Theory-based conceptual framework, 695
- Thermostat ecology, 330, 334, 337*f*
- Think-aloud data collection technique, 37, 131*b*, 438*b*, 441*b*, 448, 493, 508*b*
- emotional impact evaluation, 522–523
 - qualitative UX evaluation data
 - codiscovery, 512
 - defined, 511
 - in empirical evaluation, 512
 - management of participants, 511–512
- Thomas, J. C., 110, 338, 470
- Thomas, P., 527
- Thompson, C., 683–684
- Thomsen, A. D., 604
- Three-to-five-users rule, UX evaluation
- assumptions, 612–614
 - cumulative percentage, 611
 - marginal added detection and cost-benefit, 611–612
 - old balls-in-an-urn analogy, 610
 - participant detection rates, 610–611
 - probability function, 609–610
- Ticket Kiosk System (TKS), 133–134, 158, 181–183, 569, 574
- baseline level values, 477–478*b*

Ticket Kiosk System (TKS) (*Continued*)

- benchmark task, 461–465*b*
 - clear starting and end points, 467*b*
 - ecological validity, 471*b*
 - instructional wording, 466*b*
 - cost-importance table, 580*t*
 - elemental data note, 242*b*
 - essential use case, 205*t*, 205–206*b*
 - grouping related problems, 569–570*b*
 - ideation, 300–302*b*
 - informal quantitative testing, 563, 563*t*
 - interaction, 398
 - conceptual design, 363*f*, 363*b*
 - physical hardware requirements, 246*b*
 - priority ratios for, 577–578*b*, 577–578*t*
 - problem resolutions for, 582*t*
 - product concept statement, 91*b*
 - proposed new system, 89
 - social model entities, 217*b*
 - solution for, 574*b*
 - storyboard sketches, 367–369*b*, 369*f*
 - system support requirements, 245*b*
 - target level values, 478*b*
 - user experience
 - design requirements, 235–236*b*, 241*b*, 243*b*, 247*b*
 - goals, 457–458*b*
 - measures, 459*b*, 460*t*, 475*t*
 - metrics, 475–476*b*, 475*t*
 - user persona in, 186, 186*b*
 - user satisfaction questionnaires, 472*b*
 - user stories, 234*b*, 238–239*b*
 - work activity notes, 146–147*b*, 165–174*b*
 - extraction of, 152*b*
 - work role and user class, 456*b*
- Tidwell, J., 362
- Tohidi, M., 315
- Top-down UX design approach, 284*b*
 - abstract work activity
 - in architecture, 291*b*
 - characteristics, 289–290
 - disruptive design, 291
 - goals, 286–289

hedging against risks, 291

transition for, 290–291

in architecture, 291*b*

“T” prototypes, 409

Tractinsky, N., 520

Transaction completion slips, 721–722, 721*f*

Translation, stage of Interaction Cycle

- guidelines (*see* Cognitive affordances)

Interaction Cycle, 701

Trautmann, T., 524–525

Truss, L., 711

Tufte, E. R., 804–806

Tullis, T. S., 410, 516, 522, 618

Tungare, M., 103–104, 313–314, 344, 810–811

U

Ubiquitous computing/interaction, 102–104, 344*b*

Ullmer, B., 105, 312–313, 344

Understand Needs lifecycle activity, 47

Unit code test, 634

Unmeasured tasks, 488–489

Usability

dining experience, 9

efficiency/effectiveness, 11

engineering, 106, 606

human-computer interaction, 9–10

problem analysis, 565

testing, 17

user experience, 8, 9*f*

Usability Test Observation Form, 522

Usage research, 31, 143–145, 178, 570

Usage research data elicitation

collection of, 131–142

concepts, 119–121

existing product/system, 121

goals, 121–123

preparation for, 124–131

for product/system, 141

raw data notes, 140–141

tactical goals, 122–123

Usage research data modeling, 177–178

analysis, 179–180

- artifact model, 179*b*
 - construction, 208–210
 - defined, 207–208
- elicitation, 179–180
- flow model, 188–190*b*
 - central importance, 187
 - customer journey map, 192
 - defined, 187
 - making of, 188–190
 - for product/system, 190–192
- hybrid model, 221
- information architecture model, 214–215
- kinds of model, 178–179
- model consolidation, 221–222
- physical work environment model
 - environmental factor, 212–214
 - hardware design, 212
- social model
 - active entity, 216–217
 - concerns, 219–220
 - influences, 219–220
 - issues with, 217–219
 - for product/system, 220
 - shared workplace culture, 216
 - simplified approach, 216
 - smartphone usage, 220
- suggestions for, 179–180
- tagging model with source ID, 180
- task sequence model, 200*b*, 206
 - beyond linear model, 203
 - components, 199–200
 - defined, 197–198
 - essential use case, 204–206
 - menu planning application, 197–198*b*
 - for product/system, 198–199
 - state diagrams, 206
 - step-by-step description, 200–202
 - task interaction model, 201–202*b*
- task structure model (*see* Hierarchical task inventory (HTI))
- user persona model
 - creation, 186
 - data extraction, 185–186
 - defined, 185
 - early sketch, 187
 - Ticket Kiosk System, 186*b*
- the wheel, 178*f*
- work role model, 181–182*b*
 - defined, 181–182
 - identification, 182–183
 - mediated work role, 182
 - for product/system, 182–183
 - sub roles, 182
 - user class, 183–184
- Usage scenario, 197–199
- Usefulness, 8, 9*f*, 10, 448*b*, 481*b*, 514*b*, 702*b*, 784*b*, 831*b*
- Usefulness, Satisfaction, and Ease of Use (USE), 514, 518–519
- User-centered design (UCD), 281–282
- User class definition, 184*b*, 456, 456*b*, 458*t*
 - defined, 183
 - knowledge-based characteristics, 183–184
 - physiological characteristics, 184
 - for product/system, 184
 - skills-based characteristics, 183–184
 - work role modeling, 185
- User-created affordances, 679*b*
 - cognitive affordances, 691–692*f*
 - wake-up call to designers, 689–693
- User-defined tasks, 489
- User experience (UX), 113
 - analysts, 87
 - appropriating methods and techniques, 46–47
 - basic process components
 - design lifecycle process, 29–30, 30*f*
 - hierarchy of terms, 32, 32*t*
 - lifecycle activities, 29, 30*f*
 - lifecycle sub-activities, 30–31
 - model of, 30
 - business case, 23–25
 - characteristics, 6
 - choosing methods and techniques, 47
 - compliance phase, 276–277
 - components, 9*f*

User experience (UX) (*Continued*)

- an analogy with fine dining, 9
- emotional impact, 10–15
- meaningfulness, 16
- usability, 9–10
- usefulness, 10
- context and ecology, 7
- cost justification, 24–25
- defined, 4–7
- designer, 87
- design situations, 47
- design solutions subactivities, 35*f*
- design techniques as life skills
 - abstraction, 39–40
 - brainstorming, 43
 - critical thinking, 45
 - data/idea organization, 40–41
 - framing and reframing, 44
 - immersion, 42
 - iteration, 45
 - modeling, 41
 - note taking, 40
 - observation, 37–39
 - prototyping and envisioning, 45
 - reasoning and deduction, 44–45
 - sketching and drawing, 43
 - storytelling, 41–42
- detectives, 558
- ecology, 288*b*
- emerging desire for usability, 106
- emotional impact, 549*b*
- engagement, 14
- enticement, 14
- evaluation goals, 454*f*, 473*t*, 481
- fuss about usability, 23
- goals, 594*b*
- history, 95–106
- inspection, 547*b*, 550–551, 560, 565*b*
- interaction
 - activity-based, 19
 - Dagstuhl framework, 21
 - direct/indirect, 6
 - expanding concept, 3–4
 - localized, 18
 - system-spanning, 20–21
- intertwining lifecycle activities, 81–84, 81*b*
 - activity timing, 82–83
 - pure description, 84
- lifecycle activity, 188, 484*f*
- lifecycle process choice, 46
- mapping project parameters, 47, 48*f*
- measures, 473*t*, 477*t*
- methods, 31
- metrics, 454*f*, 474–476, 479*t*, 481
- mischaracterizations, 16
- misconceptions, 16
- peanut butter theory, 17
- practitioners, 695
- problem data management, 568–571, 570*t*
- process
 - consistency, 29
 - guiding structure, 28
 - learning, 29
 - need for, 28
 - reliability, 29
 - shared conception, 29
- professionals, 597
- rise of, 106
- service experience, 22–23
- shifting paradigms, 107–109
 - design and development, 109
 - engineering paradigm, 107–108
 - human information processing paradigm, 108
 - phenomenological paradigm, 108–109
- software engineering lifecycle, 27
- targets, 454*f*, 457*t*, 464*t*, 481
- target tables, 455–456, 456*t*
- team and software developers, 593
- techniques, 31
- totality of effects, 6–7
- usability engineering, 106
- user complaints, 23–24
- user friendliness, 16
- the Wheel, 30, 31*f*

User experience (UX) design, 7

- affordances
 - checklist, 687*f*
 - concepts, 651
 - demystifying, 652
 - issues, 652
 - kinds of, 652–654
- analytic methods, 546
- architecture, 263
- attractive designs, 13
- brand, 14–15
- conceptual design, 78*b*
- coolness and wow design, 14
- corporate culture, 14–15
- design and implementation, 637, 637*f*
- design informing model, 192
- ecology, 51*b*, 78*b*, 89*b*, 192*b*, 303*b*, 334–335*b*, 507*b*, 636*b*
- emotional perspective in, 387
- Fitts' law, 670*t*
- importance, 7–8
- inspection, 58*b*
- interdisciplinary nature, 264
- lifecycle process, 117, 118*f*, 228*f*
- marketing, 14–15
- pyramid of human needs, 268*f*
- requirements, 227
 - compose requirements statements, 241
 - degree of formality, 240
 - generic structure, 242–244, 242–244*f*
 - vs.* prototypes, 230, 230*f*
 - software and functional implications, 231
 - team selection, 241
- small-scope approach, 79–80
- state diagram, 197*b*, 206, 206*b*
- studio, 84–85, 187–190, 223–225
- task barrier, 199
- task models, 192
- top-down design, 78*b*, 630*b*, 636*b*
- universality, 263
- user stories, 228–229, 231–240
 - cleaning up, 232
 - customer written, 641

defined, 232

extrapolation requirements, 234–236

team selection, 233

ticket buyer work role, 237*f*

truth about, 641

validation of, 248–250

wants, design, 227

User experience (UX) design guidelines, 557

assessment

content and meaning of feedback, 795–804

existence of feedback, 790–792

information displays, 804–807

presentation of feedback, 792–795

system feedback, 789–790

system response, 789

history, 831–832

human memory limitations

in cognitive affordances, 755–757

kinds of memory, 715–716

short-term/working memory, 712–715

Interaction Cycle

automation issues, 786–789

structure, 717, 717*f*

system functionality, 785

system response time, 785

overall

accommodation of user differences, 823

anthropomorphism, 815–818

consistency, 810–814

helpful help, 823–824

humor, 815

reducing friction, 814–815

simplicity, 808–810

text legibility, 822

tone and psychological impact, 818

use of sound and color, 818–822

user preferences, 822

physical action

sensing objects, 778–779

user help, 779–783

scope, 709–710

translation, cognitive affordance

User experience (UX) design guidelines (*Continued*)

- avoiding errors with, 764–765
 - clarity of, 733
 - cognitive directness in, 758–760
 - complete information, 761–762
 - consistency, 739–745
 - content and meaning, 732–767
 - controlling complexity, 746–752
 - distinguishability of choices, 738–739
 - for error recovery, 765–766
 - existence, 723–725
 - legibility, 728
 - for modes, 766–767
 - noticeability, 727–728
 - precise wording, 733–738
 - presentation, 725–731
 - presentation complexity, 728
 - presentation consistency, 730–731
 - presentation timing, 729–730
 - task structure (*see* Task structure)
 - user choices and useful defaults, 752–755
 - user/usage centeredness, 762–763
 - visibility, 727
 - universality, 709–710
 - user planning
 - efficient task paths, 720
 - progress indicators, 720–721
 - system task model, 718–720
 - transaction completion slips, 721–722
 - using and interpreting, 710–711
- User experience (UX) design issues, Interaction Cycle
- assessment, 702–704
 - outcomes, 702
 - physical actions, 701–702
 - planning, 700
 - translation, 701
- User experience (UX) evaluation
- A330 airbus, 606–607
 - adapting and appropriating, 451
 - alpha and beta testing, 449–450
 - analytic evaluation methods, 441–442
 - design walkthroughs, 444–445

- early design reviews, 444–445
- automatic evaluation, 450–451
- comparison, 442
- critical incident techniques, 610*b*, 614–616
- data analysis, 566*f*
 - cost-importance analysis, 571–584
 - lifecycle activity, 562*f*
 - onion-layers effect, 585
 - process feedback, 584–585
- data collection techniques
 - qualitative, 448–449, 565–571
 - quantitative, 447–448, 561–565
- data management, 561–565, 620–621
- data types
 - objective *vs.* subjective data, 437–438
 - quantitative *vs.* qualitative data, 437
- discount methods
 - be practical, 605
 - criticism, 604
 - defined, 603
 - limitations, 604
 - rigorous methods, 604
- emotional response, 616–619
- empirical evaluation methods, 441
 - lab-based evaluation, 442–443
 - quasiempirical, 443
 - RITE, 443
- engineering evaluation, 439–441
- expert inspection, 445
- field surveys, 449–450
- goal-oriented approach, 441
- heuristics, 445, 552–553, 619*b*
 - documenting UX problems, 554
 - limitations, 556–557
 - procedure, 553–554
 - variations abound, 554–556
- inspection, 592*b*, 604*b*
- limitations, 603
- measurability, 443–444
- mechanics report
 - consistency rules, 596–597
- jargon, 597

- large amounts of qualitative data, 598–599
- personal presence, 599
- precision and specificity, 597
- reporting vocabulary, 597
- respect feelings, 598
- methods *vs.* techniques, 436
- metrics and targets, 606
- Nielsen and Molich's original heuristics, 619–620
- participants, 608–614
- physiological responses, 616
- problem descriptions, 590, 596–597
- reliability, 444, 602–606, 602*b*
- remote evaluation, 450
- reporting results
 - content types, 595–596
 - informal summative evaluation, 591, 591*b*
 - kinds of data, 591–592
 - mechanics, 596–599
 - participant anonymity, 590–591
 - qualitative evaluation, 591–592
 - quality communication, 589–590
 - report audiences, 592–595
- rigor *vs.* rapidness, 445–447
- specialized methods, 449–451
- to stakeholders, 593–594
- summative evaluation, 601*b*
 - engineering *vs.* science, 601–602
 - formal, 438–439
 - formative, 438–441
 - informal, 439
- for system, 599
- target level specification, 562*b*
- team approach, 558
- three-to-five-users rule
 - assumptions, 612–614
 - cumulative percentage, 611
 - marginal added detection and cost-benefit, 611–612
 - old balls-in-an-urn analogy, 610
 - participant detection rates, 610–611
 - probability function, 609–610
- user testing, 437
- weaknesses, 443–444
- User experience (UX) lifecycle activities, 49
 - commercial product perspectives
 - multiuser collaborative products, 62
 - single-user products, 62
 - enterprise system perspectives, 62
 - rapid methods, 58–59
 - rigor
 - complexity, 51–56
 - defined, 49–50
 - development stage, 57–58
 - domain complexity, 51–52
 - domain familiarity, 56
 - interaction complexity, 51
 - project resources, 58
 - risk aversion, 56–57
 - system complexity space, 51
 - system complexity space quadrants, 52–56
 - scope of delivery, 59–61
- User frustration, 765
- User interaction, 686*f*
- User interface (UI), 4–5, 8, 778–779, 778*f*, 780*f*
- User Interface Management Systems (UIMS), 101
- User interface satisfaction, 521
- User Interface Software and Technology (UIST), 101
- User-made artifacts, 690–693
- User needs, pyramid, 385*b*
- User participant, 508*b*, 546*b*
- User persona model, 133–134, 169, 179
 - creation, 186
 - data extraction, 185–186
 - defined, 185
 - early sketch, 187
 - Ticket Kiosk System, 186*b*
- User preferences, 822
- User research. *See* Usage research
- User satisfaction questionnaires, 10, 472–473
- User's mental model, 328, 329*f*, 330–331
- User stories, 228–229, 231–240
 - cleaning up, 232
 - defined, 232
 - extrapolation requirements, 234–236
 - team selection, 233

User stories (*Continued*)
 ticket buyer work role, 237*f*
 validation of, 248–250

User surveys, 126

User-system tasks, 136
 benchmark tasks, 488
 exploratory free “use,”, 489

Interaction Cycle
 intention shifts, 706–707
 path variations, 704–705
 secondary tasks, 706–707
 stacking, 706–707
 unmeasured tasks, 488–489
 user-defined tasks, 489

User/usage centeredness, 762–763, 763*b*

User work practice, 119–120, 138*b*

User work role model, 129*b*, 132*b*, 169, 178–179, 179*b*, 303.
See also Work role model

UTOPIA, 398–399

UX evaluation report

UX. *See* User experience (UX)

UX design studio, 223–225

UX & HCI paradigms. *See* Shifting paradigms,
 for HCI & UX

V

Van Dam, A., 100–101

Vandenbroucke, B., 683

van Kesteren, I. E. H., 603

van Oel, W., 604

Veer, G. C. v. d., 330

Velasco, C., 652

Verbal protocol, 511

Verbal self-reporting technique
 AttrakDiff questionnaires, 524–526
 questionnaires as, 523
 think-aloud techniques, 522–523

Verification and validation (V and V), 68

Vermeeren, A. P. O. S., 5, 603

Vermeulen, J., 344

Verplank, B., 101

Vertelney, L., 430

Vertical prototypes, 408–409, 408*f*

Vi, C., 652

Virginia Tech Industrial Designs Studio, Kiva,
 85, 86*f*

Virzi, R. A., 608–609

Visual bandwidth, for information display, 807,
 807*b*, 808–809*f*

Visual comp, 425*b*

Visual design, 391–392, 395*f*

Visual designer, 5

Visual sensory affordance design issues
 color, 680
 discernibility, 675–677
 distinguishability, 677–680
 noticeability, 673–675
 presentation timing, 680–681
 text legibility, 677
 visibility, 673, 674*f*

Viswanadha, K., 14

Voicemail, user reports, 531

Voting usability, 251

W

WAAD. *See* Work activity affinity diagram (WAAD)

Walkthroughs, 444–445

Wallace, V. L., 100

Wasserman, A. I., 73, 831

Waterfall lifecycle process, 32*b*, 67*f*, 110–111, 110*b*, 152*b*,
 229*b*, 626*b*, 629*f*, 635
 software engineering, 67–68, 67*f*
 verification and validation (V and V), 68

Weber, C., 28

Web evaluation questionnaires, 520

Webster, M., 821–822

Weiser, M., 102–103, 344

Welie, M. v., 362

Welsh, D., 443, 537–538

Westerman, S., 523, 618–619

Wharton, C., 696–697

Wheel UX lifecycle process, 406*f*

- Whitaker, R. D., 397–398
- White, E. A., 20, 20*f*, 255, 399
- Whiteside, J. A., 28, 254, 431, 455–456, 606
- Wickens, C. D., 269
- Wiethoff, M., 615
- Wildman, D. M., 20, 20*f*, 255, 399, 449, 512
- Williges, B. H., 508–509
- Williges, R. C., 99, 108, 438, 441–442, 444, 508–509
- Wilson, C., 556
- Winchester, W. W., III, 12
- Winegarden, C. R., 12
- Winograd, T., 102
- Wireflow prototype, 207*b*, 411–412, 412*f*, 417*f*
- Wireframes, 375–376*f*
- interaction design
 - defined, 373–374
 - path to, 372
 - prototype, 82–83*b*, 207*b*, 230*b*, 275*b*, 445*b*, 548*b*
 - creation of, 414–416
 - decks of, 418
 - defined, 410
 - design elements, 410–411
 - software tools, 432
 - static low-fidelity, 418–419
- Wixon, D. R., 28, 254, 431, 443, 508–509, 537–538, 605, 617
- “Wizard of Oz” prototyping, 431–432
- Wong, D. C., 102
- Wood, L. E., 262
- Woods, D. D., 614
- Woolrych, A., 556, 604
- Work activity affinity diagram (WAAD), 155*b*, 171–174, 288*b*, 458
- bottom-up process
 - growing labels with groups, 158–160, 159*f*
 - labels for groups of notes, 158
 - posting work activity notes, 157–158
 - splitting large groups, 160
 - group splitting, 160*b*
 - highlights display creation, 165–174
 - observations, 174
 - organizing groups into hierarchy, 163–164
 - print work activity notes, 154
 - team working, 162*f*
 - technology support, 162–163
 - topical label, 159*f*
 - user work roles, 157*f*
 - work space, 156
- Work activity instances, 285
- Work activity note, 141*b*, 144, 182*b*, 233*b*, 244*b*, 281*b*, 308*b*, 457*b*
- data analysis subactivities
 - additional information, 150
 - brief and concise, 147–148
 - into categories, 151
 - complete, 148
 - context retain, 148–149
 - data sources, 150–151
 - elemental note, 146–147
 - extraction, 151–153
 - handwritten/typed, 146
 - organization, 154–155
 - for product/system, 154
 - with source ID, 150–151*b*
 - synthesization, 147*b*
 - TKS, 146–147*b*
 - user stories and requirements, 151–152
- Work artifact model, 123*b*, 207–208*b*, 208*f*, 307*b*
- Work domain, 119
- Work environment model, 179, 211*b*
- environmental factor, 212–214
 - hardware design, 212
- Workflow synchronization, agile SE
- communication during, 646–647
 - continuous delivery, 646
 - dove-tailed work activities, 643–645
 - early delivery value, 645–646
 - planning across iterations, 646–647
 - regression testing, 646
- Working memory, 712–715, 768

Work practice, 180*b*, 285

Work role model, 181–182*b*

- defined, 181–182
- hierarchical task inventory, 195
- mediated work role, 182
- for product/system, 182–183
- subroles, 182
- user class, 183–184 (*see also* User class definition)

Wright, P. C. K., 131, 428, 769

Y

Ye, S. X., 104

Young, R. R. M., 99, 231

Z

Zhang, P., 11–12, 555–556

Zhang, Z., 555–556

Zheng, W., 103–104, 313–314

Zieniewicz, M. J., 102